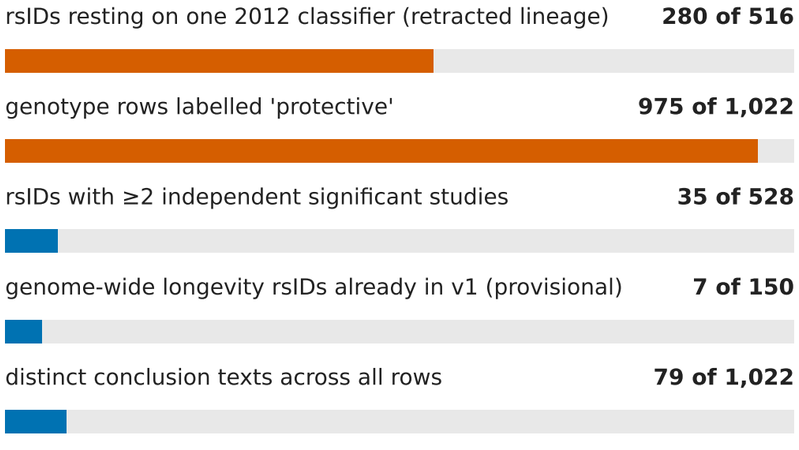
longevitymap v2: from a frozen candidate-gene list to evidence-tiered, mechanism-first longevity genetics

Requirements summary, 2026-11-03. Numbers come from the v1 audit and the literature tracks; GWAS-coverage and tier counts are provisional until curation.

a v1 rests on one overfit study and misses modern loci



b One tier gates scoring and default visibility

A — Well established (11 lead loci)
genome-wide in ≥100k lives or ≥3k cases above the 90th percentile; replicated

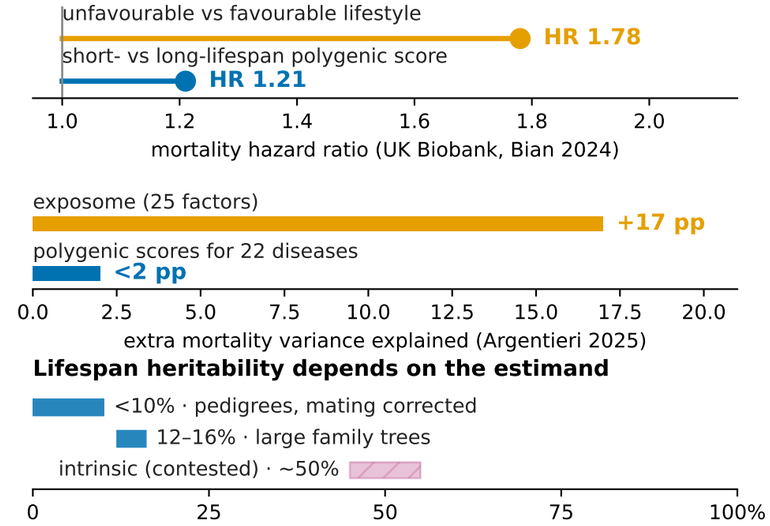
B — Likely (82, to prune)
genome-wide locus, but one study, a tag SNP, or ancestry-specific replication only

C — Early evidence (39)
suggestive, small replication or preprint; research view only

D — Not supported (1 + v1 legacy)
candidate-only, contradicted, meta-null, retracted or overfit source (v1 legacy)

Reported separately, never folded into the tier: mechanism, pleiotropy, age of action, outcome (healthspan / lifespan / very old age), actionability, context, causal confidence, animal support.

c Lifestyle outweighs inherited polygenic risk



d Variants are grouped by how they act, because most lifespan signal runs through a disease or behaviour

Heart and vessels
LPA · LDLR · 9p21 region
CELSR2/SORT1 · PCSK9
preventable: a doctor can measure Lp(a), LDL-C, BP

Brain and memory
APOE ε2 / ε3 / ε4 ★
scored once, as a pair
descriptive; opt-in, Alzheimer's implication stated plainly

Smoking and lungs
CHRNA3/5
(nicotine receptor genes)
modifiable: the behaviour, not the variant, can change

Immune system
HLA-DQA1/DRB1 · SH2B3 ⇄
IL6R · IL6 (Chinese cohorts)
trade-offs: less autoimmunity and CVD, more cancer/infection

Blood sugar and weight
TCF7L2
FTO / MC4R (via BMI)
modifiable: weight and glucose are measurable, changeable

Skin and sun
IRF4 · TYR · DEF8
(pigmentation; healthspan)
modifiable: acts through skin cancer; sun protection

Ageing in general
FOXO3 · ZW10/USP28
FGD6 ★ · HTT ★
descriptive: no single disease route (FURIN, KCNK3 pending)

Ageing markers
TERT ★ / TERC · clocks
loss of Y · mtDNA copies
not scored: a marker tendency, not a longer-life allele

★ also supported in animal studies (context only; never raises a tier) ⇄ trade-off: shown as 'may lower / may raise', never as a good/bad colour Examples are seed leads

e One report, three layers of detail

Citizen scientist (default view)
- the result in plain words, about you
- copies of the longer-life version (0, 1, 2)
- certainty word, what it acts through
- 'may lower / may raise' for trade-offs
- what a doctor can measure instead
- ancestry caveat when it applies

Geneticist (More details fold)
- rsID, GRCh38 position, ref/alt, zygosity
- effect allele, effect size + unit, CI, p
- phenotype class, cohort N and ancestry
- fine-mapping and effector-gene confidence

Ageing researcher (research view)
- tiers C/D, negative and contradicted evidence
- legacy LongevityMap rows (frozen 2017)
- full pleiotropy list, hallmark tags
- animal-model support

f Key principles

1 Evidence first: only genome-wide, replicated signals are scored; everything else is research view.
2 One lead variant per locus; LD proxies and APOE-region tags are never counted twice.
3 A lead SNP 'marks a region' unless fine-mapping makes it likely causal; name the gene only when confident.
4 Say what it acts through, at what age, and on healthy years or length of life; state trade-offs both ways.
5 Every genotype gets a row; the most common genotype is the neutral reference.
6 Ancestry, sex and age limits travel with the finding; no transfer of European effects to other ancestries.
7 Every claim has a receipt: PMID, effect allele, effect size with unit, and a located quote.

Never
× No summed 'longevity score' or net weight from single SNPs
× No predicted lifespan or genotype-based 'biological age'
× No red/green good-bad colouring, least of all on trade-offs
× No odds ratios for citizens without plain-language context
× No 'gene that controls ageing' wording
× No animal result raising a human tier or weight